



A high pressure shock tube study of *n*-propylbenzene oxidation and its comparison with *n*-butylbenzene

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ABSTRACT

Ignition delay times have been measured for mixtures of *n*-propylbenzene in air ($\approx 21\% \text{ O}_2$, $\approx 79\% \text{ N}_2$) at equivalence ratios of 0.29, 0.48, 0.96 and 1.92 and at reflected shock pressures of 1, 10 and 30 atm in a heated high-pressure shock tube over a wide temperature range (1000–1600 K). The effects of reflected shock pressure and of equivalence ratio on ignition delay time were determined and common trends highlighted. Simulations were carried out using the *n*-propylbenzene sub-mechanism contained in an *n*-butylbenzene reaction mechanism available in the literature. This kinetic model was improved by including pressure dependent reactions which were not in place previously and the addition of the NUI Galway C₀–C₄ sub-mechanism. These simulations showed very good agreement with the experimental data. Additionally a comparison is made with experimental data previously obtained and published for *n*-butylbenzene over the same range of conditions and common trends are highlighted.

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1. Introduction

Due to dwindling fossil fuel supplies it is important to use remaining supplies efficiently. Today's transport fuels contain a large proportion of aromatic and alkyl-aromatic compounds, petrol contains $\approx 30\%$ aromatic compounds while diesel contains $\approx 9.2\%$ alkylaromatic compounds [1]. There is an increased awareness of the environmental impact resulting from the burning of these fossil fuels. The key to being able to improve the efficiency, performance and emissions of combustion systems is a better understanding of the chemistry involved over the wide range of temperatures and pressures relevant to combustor operation. Therefore, there is much interest in understanding the oxidation of hydrocarbons. To date there has been little research published on the combustion characteristics of large *n*-alkylbenzenes, but the aromatic compound, benzene [2–12], mono-substituted *n*-alkylbenzenes like toluene [4,12–19], di-substituted alkylbenzenes like xylenes [20–25], and to a lesser extent larger alkylbenzenes have been investigated [17,26–32].

The combustion of benzene has been studied and described using detailed chemical kinetic reaction mechanisms in

low-pressure laminar flame composition profiles [2,3], flame speed measurements [4], well-stirred reactors by Chai and Pfefferle [5] and by Ristori and Dagaut [6], flow reactors [7–10] and shock tubes [11,12].

Methyl benzene or toluene has also been the subject of a series of studies, in a flow reactor [13,14], a jet-stirred reactor by Dagaut et al. [15], shock tubes [12,16], a rapid compression machine by Roubaud et al. [17] and diffusion and laminar flames [4,18]. The oxidation of toluene was also simulated by Lindstedt and Maurice [19] using experimental data derived from diffusion flames, plug flow reactors, shock tubes and premixed flames.

Xylene has also been studied in the past but to a lesser extent. Battin-Leclerc et al. studied three isomers of xylene in a high pressure shock tube [20]. *o*-Xylene was also studied along with *o*-ethyltoluene and *n*-butylbenzene in a rapid compression machine (RCM) by Roubaud et al. [21]. Other xylene studies include those by Johnson and Farrell [22], Gail and Dagaut [23], Shen and Oehlschlaeger [24] and Gudiella et al. [25].

The alkyl fragments on aromatic compounds in transport fuels can range from C₁ (methyl) all the way up to C₁₆ (hexadecyl) or larger carbon side-chains with branched side chains common; however there is an obvious knowledge gap concerning the combustion of the alkyl benzenes with larger alkane fragments.

n-Propylbenzene has been studied previously [17,26–28], with these works focusing on temperatures below 1200 K. Litzinger

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et al. [26] used a plug-flow reactor to study the oxidation of *n*-propylbenzene at 1 atm pressure, at a temperature of approximately 1060 K and at equivalence ratios of 0.65, 1.0 and 1.5 highly diluted in nitrogen (99%).

Roubaud et al. [17] carried out rapid compression machine experiments on eleven different alkylbenzenes (including *n*-propylbenzene) in the lower temperature region (600–900 K), at compressed pressures of up to 25 bar for stoichiometric mixtures in air.

Dagaut et al. [27] carried out experiments in a jet-stirred reactor (JSR) at atmospheric pressure over a temperature range of 900–1250 K at atmospheric pressure and at equivalence ratios of 0.5, 1.0 and 1.5 highly diluted in nitrogen (≥ 50 ppm of O_2 and H_2O ; ≥ 1000 ppm of Ar; ≥ 5 ppm of H_2). These experiments provided concentration versus temperature profiles of 23 different species. In addition, they developed a kinetic reaction mechanism for the oxidation of *n*-propylbenzene containing 124 species and 985 reactions.

A recent study by Gudiya and Brezinsky [28] produced high-pressure single-pulse shock tube speciation data obtained at shock pressures of 25 and 50 atm, at temperatures between 838–1669 K and equivalence ratios between 0.5–1.9. Species concentration versus temperature data for a variety of stable species was reported. A kinetic model was also used to simulate the data in this study.

A detailed chemical kinetic mechanism has been used to simulate our data. This mechanism is mostly derived from that published by Dagaut et al. [27] for *n*-propylbenzene oxidation but which was subsequently modified by that group in their study of *n*-butylbenzene oxidation [29] in a jet-stirred reactor. The *n*-butylbenzene mechanism was also derived from that produced by Mati et al. [30] to describe the oxidation of a diesel fuel. Mati et al. studied a surrogate for diesel fuel using *n*-hexadecane, *n*-propylcyclohexane, *n*-propylbenzene, iso-octane and 1-methylnaphthalene in a jet-stirred reactor at pressures from 1–10 atm over a temperature range of 800–1400 K and at equivalence ratios in the range 0.5 and 2.0. A model was also produced for this surrogate [30]. The experimental data in *this study* has been simulated using the *n*-propylbenzene sub-mechanism from the Diévert and Dagaut [29] mechanism. Another study of *n*-butylbenzene carried out by Pousse et al. [31] also included an *n*-propylbenzene sub-mechanism but this mechanism was not validated against *n*-propylbenzene data. As the mechanism of Diévert et al. had previously been employed to simulate *n*-propylbenzene oxidation we chose to test and employ this mechanism in our study.

This study aims to bridge a gap between the already studied aromatic and alkyl aromatic compounds and longer-chain alkyl aromatic compounds by investigating the kinetics of *n*-propylbenzene oxidation which in turn can be related to the larger alkylbenzenes discussed above. In particular, it will provide ignition delay time measurements in the high temperature regime (1000–1600 K) for *n*-propylbenzene measured in the heated high pressure shock tube (HPST) facility at NUI Galway and compare these ignition delay times with a previously published chemical kinetic mechanism.

Moreover, the experimental data obtained in *this study* for *n*-propylbenzene oxidation will be compared to data also obtained for *n*-butylbenzene oxidation under similar conditions and has been submitted for publication in a separate study [32].

2. Experimental

2.1. Apparatus

Ignition delay times were measured in a high-pressure shock tube consisting of a stainless steel tube of 8.76 m in length, with an internal diameter of 6.3 cm. A double-diaphragm section

divides the shock tube into a 3 m long driver section and a 5.73 m driven section. Polyethylene terephthalate films (KATCO) were used as diaphragms in all experiments, where the thickness of the diaphragm material was chosen depending on the desired final shock pressure and varied from 75–500 μm . The driver gas used was helium (99.99% pure; BOC). The operational pressure limit of the shock tube is approximately 60 bar. The diagnostic system involves four pressure transducers, where the velocity of the incident shock wave was measured at three locations separated by known distances with the shock velocity extrapolated to the end-wall. The pressure at the endwall was monitored using a pressure transducer (PCB, 113A24). The incident shock velocity at the end-wall was used to calculate the temperature and pressure of the mixtures behind the reflected shock wave using the equilibrium program Gaseq [33] (see Fig. 1).

The ignition delay time was defined as the interval between the rise in pressure due to the arrival of the shock wave at the endwall and the maximum rate of rise of the pressure signal, Fig. 2. Pressure traces were obtained using a Sigma digital oscilloscope (Sigma, 90–4). Any pressure profiles exhibiting significant pre-ignition pressure rise ($\geq 5\%/ms$) were excluded from the study as this has been shown to cause inaccurate measurements of ignition times in a study carried out by Shen et al. [34].

Estimated uncertainty limits of the measurements are ± 15 K in reflected shock temperature, T_5 , $\pm 15\%$ in ignition delay time, τ , and $\pm 10\%$ in mixture concentration.

2.1.1. Heating system

A heating system was designed and installed on the driven section and manifold of the shock tube to ensure that *n*-propylbenzene, which has a saturation vapor pressure of 0.5 kPa at 25 °C [35], was present in the gaseous phase for the duration of the experiment and that no condensation of the fuel occurred.

Three layers of insulation tape (Zetex 1000) were placed along the length of the driven section of the shock tube and part of the high pressure section; sensors connected to a digital output (Omega, HH804) were placed under the insulation tape at regular intervals to measure the temperatures along the length of the driven section.

Double-stranded heating tape (Flexelec, 1250 W) was wrapped over the insulation tape with thermocouples positioned underneath at regular intervals. The heating tapes were connected to thermostats (Flexelec, 16 amp) which were controlled by the thermocouples. A final layer of insulation was wrapped over the heating tape to ensure a minimum loss of heat to the atmosphere. This heating system maintained temperatures within ± 3.5 °C of the preset value, 100 °C along the driver and driven section of the shock tube. This temperature was chosen to ensure that the vapor pressure of *n*-propylbenzene was well below its saturation vapor pressure and hence was always in the gaseous phase.

The heating system was extended to the manifold where three rope heaters (Omega, FGR) and one single stranded heating tape

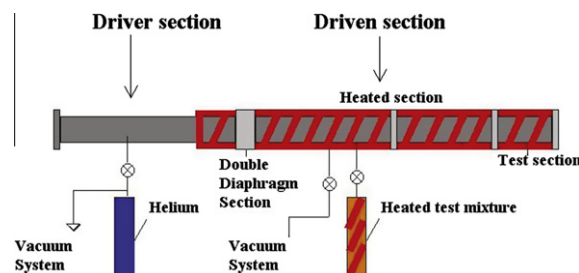


Fig. 1. Schematic of the high pressure shock tube.

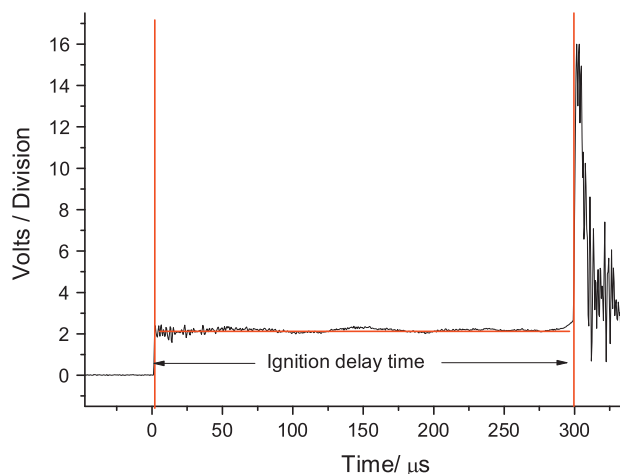


Fig. 2. Typical pressure trace; $\phi = 0.96$, $p_5 = 29.65$ atm, $T_5 = 1132$ K, Ignition delay time = 295 μ s; Fuel: *n*-propylbenzene.

(Flexelec, 1250 W) were wrapped directly onto the surface of the manifold. The ropes and heating tape were attached to four separate thermostats (Vulcanic 30633, 2 amp output), controlled by thermocouples. As before the sensors measured the temperature at different locations. The heating ropes and tape on the manifold were finally wrapped with insulation tape. The heating system design used on the driven section of the shock tube was repeated to heat the two mixing tanks used herein, with three thermostats (Flexelec, 16 amp) used per tank.

2.2. Mixture preparation

Two methods of mixture preparation were employed. These methods were:

1. Direct injection of the fuel using an injection port on the top of the heated mixing tanks using a 5 ml gas-tight syringe (SGE Analytical Science, 008760). A weighed amount of fuel was added and the partial pressure of fuel was measured using an MKS pressure transducer and digital readout followed by the addition of oxygen and nitrogen to a final known pressure.
2. Evaporation of the de-gassed liquid fuel (freeze–pump–thaw) to the required partial pressure followed by the addition of oxygen and nitrogen to the desired final pressure.

Both procedures used to generate the mixtures gave essentially the same results. For convenience the direct injection technique was utilized for the majority of the experiments.

n-Propylbenzene was supplied by Tokyo Chemicals Ltd. ($\geq 99\%$ purity GC grade). Oxygen ($\geq 99.5\%$ purity) and nitrogen ($\geq 99.998\%$ purity) were supplied by BOC Ltd. Care was taken that the maximum fuel pressure utilized was well below the saturation vapor pressure at the operating temperature of the manifold and shock tube. Table 1 describes the composition of fuel to be used in the study. Equivalence ratios of 0.3, 0.5, 1.0 and 2.0 were intended but mistakenly, to calculate the appropriate oxygen concentration a stoichiometric coefficient of 12.5 times the fuel concentration was used instead of 12.0.

3. Results and discussion

3.1. Shock tube validation

The shock tube and heating system were validated by comparison with data obtained by Davidson and Hanson [36] for

Table 1

Composition of *n*-propylbenzene mixtures studied.

ϕ	% Fuel	% O ₂	% N ₂
1.92	3.25	20.32	76.43
0.96	1.65	20.65	77.70
0.48	0.83	20.83	78.34
0.29	0.50	20.89	78.61

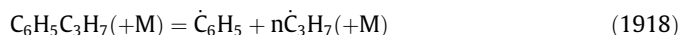
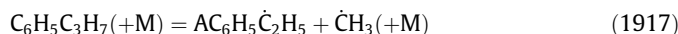
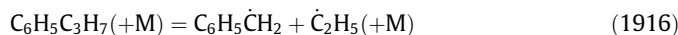
n-heptane oxidation in air at an equivalence ratio (ϕ) of 1.0 and reflected shock pressure of 10 atm.

Varying the mixing tank, manifold and shock tube temperature from 25 °C to a maximum of 125 °C had no impact on the measured ignition delay times, which were in very good agreement with the Davidson data [36], Fig. 3.

3.2. Chemical kinetic model

CHEMKIN-PRO [37] was used to simulate the shock tube experiments assuming adiabatic, constant volume conditions with no inhomogeneities. The *n*-propylbenzene sub-mechanism used formed part of an *n*-butylbenzene mechanism published by Diévar and Dagaut [29] composing 404 species and 2210 reactions.

Initial simulations carried out using this mechanism, see long-dash line in Fig. 4, showed that it was consistently slow compared to the data. It contained no pressure dependent rates for the unimolecular decomposition reactions which are important for these experiments and in this temperature regime. The rate constants for these reactions were calculated based on microscopic reversibility using an estimate of the rate constant in the reverse direction for radical–radical recombination. For all reactions (R1916–R1918) a rate constant of $1.2 \times 10^{13} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ was used for the recombination of two radicals to form *n*-propylbenzene.



A chemical activation formulation based on Quantum Rice–Ramsperger–Kassel theory, as described by Dean [38,39] was then used together with the high-pressure limit expressions to develop

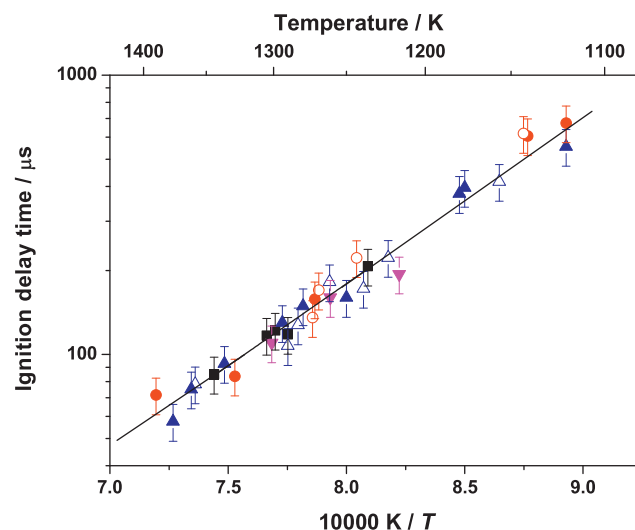


Fig. 3. *n*-Heptane validation; 10 atm; $\phi = 1.0$, ■ – Davidson et al. data, ● – room temperature; evaporation method, ○ – room temperature; direct injection, ▲ – 75 °C; direct injection, △ – 125 °C; direct injection, ▼ – 100 °C; direct injection method. — : Linear fit of data with $R^2 = 0.97146$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

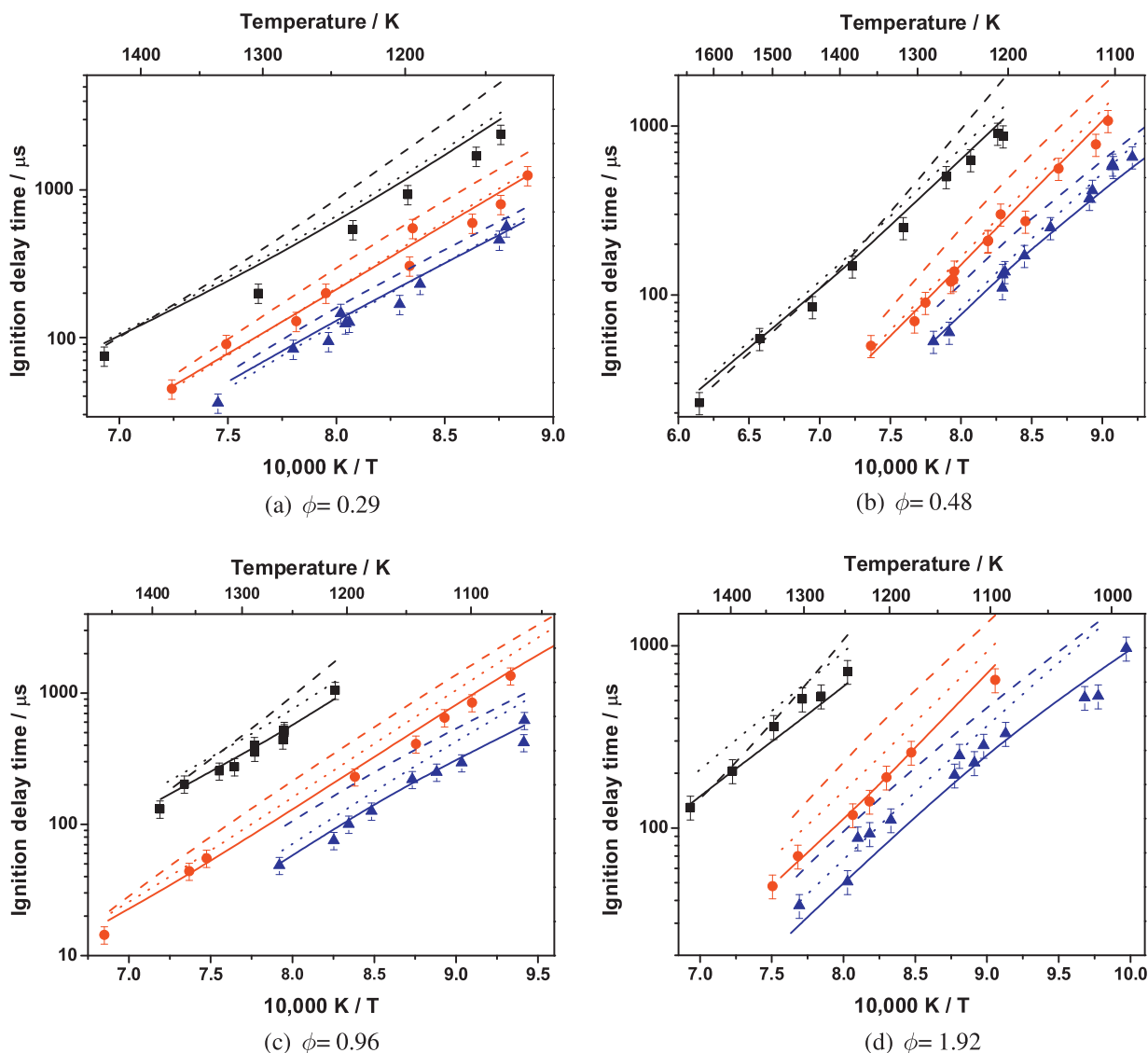


Fig. 4. Effect of varying reflected shock pressure on *n*-propylbenzene reactivity; ■ – 1 atm, ● – 10 atm, ▲ – 30 atm. Simulation; – – – Diévert mechanism, ··· Diévert mechanism + unimolecular falloff, — Diévert mechanism + unimolecular falloff + C₀–C₄. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

pressure-dependent rate constants which were fitted to a nine-parameter Troe formalism [40]. These calculated rates can be found in Table 3 where all changes made to the *n*-propylbenzene mechanism have been detailed.

In addition to these reactions, the C₀–C₄ sub-mechanism forming part of the Diévert and Dagaut [29] mechanism was replaced by one under development at NUI Galway. This sub-mechanism is based upon previously published work [41–45]. The mechanism with these new rates captured successfully the data produced in this study for all data sets, Fig. 4a–d. The current mechanism shows a marked improvement in relation to the original mechanism.

Additionally rate constants for the abstraction of a secondary allylic hydrogen atom, forming 1-phenyl-1-propyl radicals, C₆H₅C₃H₆–C (pbzjC) in the case of propylbenzene and 1-phenyl-1-butyl radicals, C₆H₅C₄H₈–D in the case of butylbenzene have been modified based on estimates in this study using analogies with primary allylic hydrogen atoms in propene and the variation/increase in rate constant observed when comparing abstraction of a secondary hydrogen atom compared to a primary hydrogen atom abstraction. The rate constants used are provided in Tables 3 and 4. In our mechanism abstraction of these secondary

allylic hydrogen atoms are favored for both alkylbenzenes and the subsequent decomposition of the radical formed leads to the formation of styrene and a methyl radical in the case of propylbenzene and an ethyl radical for butylbenzene. The latest version of this updated mechanism and thermodynamic data are included as Supplementary Material.

3.3. *n*-Propylbenzene oxidation results

The results obtained for the oxidation of *n*-propylbenzene are described below. Error bars of ±15% of ignition delay have been included in all plots to account for the uncertainty of measurement of these results. A complete glossary of all the shortened species names required for CHEMKIN-PRO and all experimental data can be found in the Supplementary Material.

3.3.1. Effect of pressure on ignition delay time

The effect of pressure on ignition delay times was studied at four equivalence ratios in air (0.29, 0.48, 0.96 and 1.92), Fig. 4. Ignition delay times of *n*-propylbenzene oxidation versus inverse temperature at an equivalence ratio of 0.29 and at pressures of

approximately 1, 10 and 30 atm are shown in Fig. 4a. At these conditions, reactivity increases with increasing pressure, that is shorter ignition delay times were measured for higher pressures. The same trends are observed for fuel/air mixtures at equivalence ratios of 0.48, 0.96 and 1.92 at the same reflected shock pressures, Fig. 4b–d. This indicates a consistent trend of reactivity of *n*-propylbenzene with pressure throughout this study. These results are typical of the effect of pressure on ignition delay time [46].

While the general trend of an increase in pressure resulting in an increase in reactivity is consistent throughout the range of equivalence ratios, the extent of this increase is much smaller at $\phi = 0.29$ compared to the variation over the 1, 10 and 30 atm pressure data observed at $\phi = 1.96$.

3.3.2. Effect of equivalence ratio on ignition delay time

The effect of equivalence ratio on ignition delay times was determined by measuring ignition delay times for four fuel/air mixture compositions at 1, 10 and 30 atm, Fig. 5. The effect of equivalence ratio at a reflected shock pressure of 1 atm at equivalence ratios of 0.29, 0.48, 0.96 and 1.92 is shown in Fig. 5a. Essentially this Figure shows that under these conditions there is very

little dependence of fuel reactivity on equivalence ratio over the temperature range 1100–1450 K. However, if one looks closely enough at the relative slopes of the data it can be concluded that the fuel-rich mixtures ignite more slowly at the highest temperatures compared to fuel-lean mixtures and as the temperature decreases this difference becomes less pronounced. Due to this trend it is expected that there will be a crossover in reactivity as the temperature becomes even lower. This is because at lower temperatures chain branching depends upon fuel concentration. The reactivity of all equivalence ratios at 1 atm is similar and it is only in the higher temperature range that the trends described above can be easily observed.

The effect of equivalence ratio is more pronounced at 10 and 30 atm. At 10 atm all mixtures have similar reactivities at the highest temperatures studied as there is a convergence of all equivalence ratio data in the region of 1300–1400 K while fuel-rich mixtures ignite faster as the temperatures decrease for the reasons outlined above, Fig. 5b. The trend in reactivity is the similar for 30 atm although the temperature in which the convergence occurs cannot be clearly seen due to the high reactivity of the fuel, Fig. 5c. Fuel-rich mixtures ignite faster than fuel-lean mixtures for the

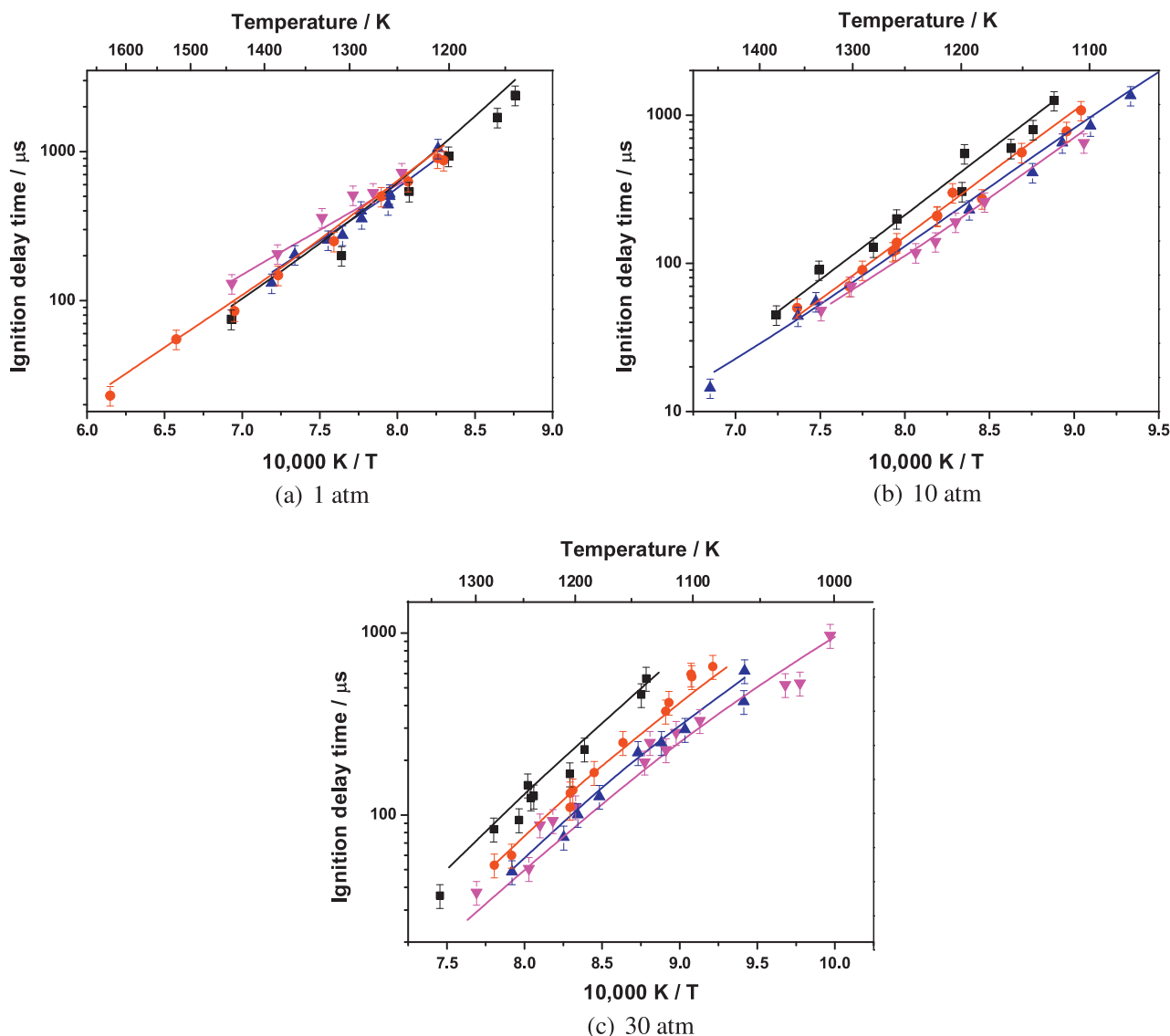


Fig. 5. Effect of varying equivalence ratio (ϕ) on *n*-propylbenzene reactivity; ■ – $\phi = 0.29$, ● – $\phi = 0.48$, ▲ – $\phi = 0.96$, ▼ – $\phi = 1.92$. Simulation; — Diévert model + unimolecular falloff + C_0 – C_4 . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

entire range. At approximately 1330 K the data at equivalence ratios 1.92 and 0.96 indicate a convergence where reactivity is similar while the trends of the data at leaner conditions indicate that a convergence will occur at a higher temperature than is available to study for this pressure. The reason for this is that the fuel is so reactive it is impossible to accurately determine the ignition delay time as it becomes too short ($\leq 30 \mu\text{s}$). From here it can be seen that the crossover from high- to low-temperature chemistry occurs at varying temperatures at different reflected shock pressures. As the pressure increases this crossover occurs at higher temperatures.

3.4. Sensitivity analysis and reaction pathway analysis

A series of sensitivity and reaction pathway analyses were carried out to investigate reaction sensitivity to ignition delay time for *n*-propylbenzene oxidation as a function of temperature, pressure and equivalence ratio and these can be found in the [Supplementary Material](#) accompanying this paper. The effect of temperature is discussed in detail below.

3.4.1. Sensitivity analysis

Sensitivity analyses were carried out using a brute force analysis which increases and decreases both the forward and reverse rate constants by a factor of two using an automated code developed in-house, with sensitivities expressed using the formula:

$$\ln S = \frac{\ln(\tau_+/\tau_-)/\ln(k_+/k_-)}{\ln(k_+/k_-)} = \frac{\ln(\tau_+/\tau_-)}{\ln(2/0.5)}$$

Sensitivity coefficients showing the effect of temperature on the oxidation of *n*-propylbenzene is shown in Fig. 6 in which a positive sensitivity value indicates an inhibiting reaction while a negative sensitivity value relates to a reaction promoting reactivity.

At 1000 K in particular, abstraction of a hydrogen atom by molecular oxygen to form a pbzjC radical and a hydroperoxyl radical (R1924), Fig. 7c, is the most important reaction promoting reactivity as it is the main initiation reaction at this temperature. This leads to the formation of one HO_2 radical initially and a pbzjC radical which further decomposes to styrene and a methyl radical. Thus, for each molecule of fuel that reacts with molecular oxygen two relatively reactive radicals, HO_2 and $\dot{\text{C}}\text{H}_3$, are formed. Reactions involving the hydroperoxyl radical have a strong promoting effect on the system as can be seen by the reaction of the benzyl radical with hydroperoxyl radicals to form a methoxy-phenyl

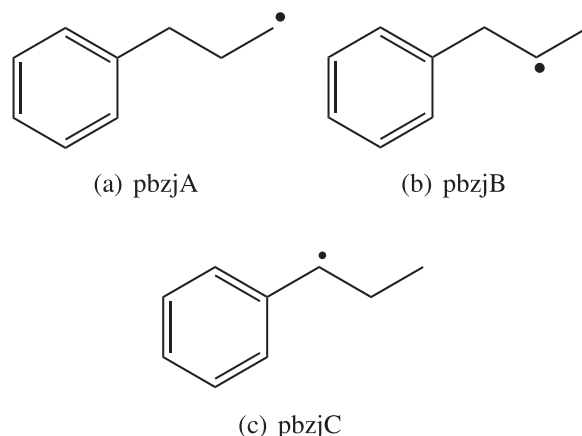
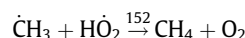
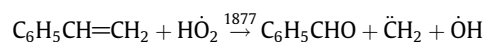
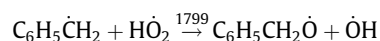
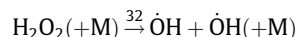
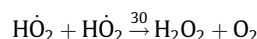


Fig. 7. Structure of *n*-propylbenzene radicals.

radical and an hydroxyl radical (R1799) which helps promote reactivity as it abstracts hydrogen atoms from a stable molecule.



A reaction which inhibits reactivity involves the self-reaction of hydroperoxyl radical to generate hydrogen peroxide and molecular oxygen (R30). This reaction consumes two hydroperoxyl radicals, producing only one hydrogen peroxide molecule which subsequently decomposes (R32) into two hydroxyl radicals (chain propagation). This reaction thus competes with other $\text{RH} + \dot{\text{H}}\text{O}_2 = \dot{\text{R}} + \text{H}_2\text{O}_2$ reactions in which only one radical generates one hydrogen peroxide molecule. Conversely a reaction promoting reactivity is the reaction between styrene ($\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$), and an hydroperoxyl radical to form benzaldehyde, methylene and an hydroxyl radical (R1877). The termination reaction of a methyl radical and an hydroperoxyl radical to form methane and molecular oxygen (R152) is one of the most important reactions inhibiting reactivity due to the competition it provides with the above reactions by consuming $\dot{\text{H}}\text{O}_2$ radicals.

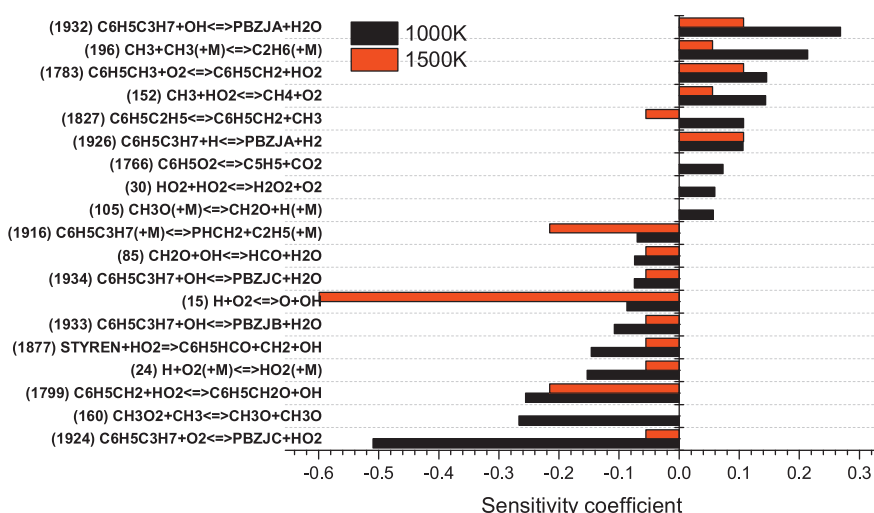
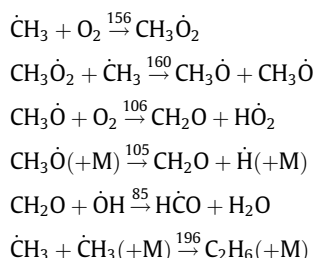


Fig. 6. Sensitivity coefficients showing the effect of temperature on *n*-propylbenzene ignition delay time, $\phi = 0.96$, $p_5 = 10 \text{ atm}$.

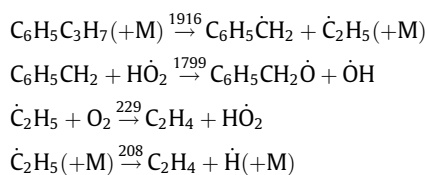


Methyl radicals react with molecular oxygen to produce methylperoxy radicals (R156), and these react with methyl radicals to produce two methoxy radicals (R160), which react with molecular oxygen to form formaldehyde and a hydroperoxyl radical (R106). This reaction is an important reaction promoting reactivity due to the formation of a hydroperoxyl (HO_2) radical which can ultimately result in two hydroxyl ($\dot{\text{O}}\text{H}$) radicals in addition to formaldehyde reacting with a hydroxyl radical to form a formyl ($\text{H}\dot{\text{C}}\text{O}$) radical and water (R85) which also promotes reactivity. Competing with this reaction is the decomposition of a methoxy radical to form formaldehyde and a hydrogen atom (R105). This reaction is an important reaction inhibiting reactivity as it competes with the above reaction which promotes reactivity. Another reaction involving the methyl radical which inhibits reactivity is the recombination of two methyl radicals to form ethane. This reaction combines two reactive radicals to form a stable species and as such inhibits reactivity (R196).

Hydrogen atom abstraction from the fuel by $\dot{\text{O}}\text{H}$ radicals, to form a $\text{C}_6\text{H}_5\text{C}_3\text{H}_6\text{-A}$ radical (pbzjA) and water (R1932), is the most important reaction inhibiting reactivity at 1000 K. This is because it competes with the hydrogen atom abstraction reactions by $\dot{\text{O}}\text{H}$ radicals which produce pbzjB (R1933) and pbzjC (R1934) radicals both of which strongly promote reactivity. This is because the products of the decomposition of pbzjA radical, benzyl radical and ethylene, are less reactive than the products formed by the decomposition of either pbzjB radical, which decomposes to form a phenyl radical and propylene, or pbzjC radicals.

The third most inhibiting reaction is one in which toluene reacts with molecular oxygen to form a benzyl radical and a hydroperoxyl radical (R1783). This reaction competes for molecular oxygen with the main initiation reaction of *n*-propylbenzene with molecular oxygen leading to the formation of a pbzjC and a hydroperoxyl radical, as discussed previously.

At 1500 K the dominant reaction of the fuel promoting reactivity is the unimolecular decomposition of *n*-propylbenzene to form $\text{C}_6\text{H}_5\dot{\text{C}}\text{H}_2$ (benzyl) and C_2H_5 (ethyl) radicals (R1916). This reaction promotes reactivity as it leads to the formation of ethyl radicals which can react with molecular oxygen to form ethylene and an hydroperoxyl (HO_2) radical (R229). Additionally the benzyl radical reacts with the hydroperoxyl radical to form $\text{C}_6\text{H}_5\text{CH}_2\dot{\text{O}}$ radical and a hydroxyl radical which helps promote reactivity as it abstracts hydrogen atoms from a stable molecule (R1799). This reaction does not occur to the same extent at lower temperatures, as there is not enough energy available to break this C–C bond, and as such does not promote reactivity as much as at 1500 K.

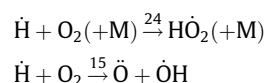


Interestingly, ethyl radical decomposition to ethylene and a hydrogen atom (R208) inhibits reactivity at 1500 K (sensitivity coefficient of +0.056) as it competes with the reaction of ethyl radical with molecular oxygen (sensitivity coefficient of –0.056)

which also produces ethylene and a hydroperoxyl radical (R229). At 1000 K hydroperoxyl radical chemistry more strongly promotes reactivity compared to 1500 K. At both temperatures, ethylene reacts with hydroxyl radicals to form vinyl radicals and water.

The reaction of fuel molecules with hydrogen atoms compete with hydrogen atom reactions with molecular oxygen in the high temperature chain branching reaction $\dot{\text{H}} + \text{O}_2 = \dot{\text{O}} + \dot{\text{O}}\text{H}$ (R15) and as such all of these reactions inhibit reactivity. Generally at 1000 K the sensitivity of the system to hydrogen atom abstraction is stronger than at 1500 K because of the competition for fuel molecules introduced through the increased importance of unimolecular decomposition.

At intermediate and low temperatures the reaction of hydrogen atoms with molecular oxygen results in the formation of a hydroperoxyl radical (R24). At high temperature the reaction forms $\dot{\text{O}} + \dot{\text{O}}\text{H}$ and is the dominant high temperature chain branching reaction. This is shown in Fig. 6 as at 1000 K the system is far less sensitive to $\dot{\text{H}} + \text{O}_2 = \dot{\text{O}} + \dot{\text{O}}\text{H}$ than it is at 1500 K and it is the most sensitive reaction at 1500 K.



It can be seen that at these conditions the temperature plays a key role in which reactions are the most sensitive. At lower temperatures the majority of the sensitive reactions related to the fuel are hydrogen atom abstractions while at 1500 K the most sensitive reactions are due to unimolecular decomposition of the fuel and subsequent reactions of the daughter radicals. This discussion of the important reaction channels of the small radicals ($\dot{\text{C}}\text{H}_3$, $\dot{\text{C}}_2\text{H}_5$, etc.) are supported by rate of production analyses. However, we discuss only the primary fuel reactions and those important to its daughter radicals in our reaction path analyses below.

3.4.2. Reaction pathway analyses

3.4.2.1. Low–intermediate temperature (1000 K). A reaction path analysis was carried out at 15% fuel consumption at 1000 K at an equivalence ratio of 0.96 and a compressed pressure of 10 atm, Fig. 8. From these analyses it can be seen that at these conditions low to intermediate temperature chemistry is occurring as 75.8% of the fuel decomposes via hydrogen atom abstraction. Hydrogen atoms can be abstracted from one of three positions on the alkyl side-chain. These reactions result in a series of propylbenzene radicals (pbzjc, pbzjb and pbzjC), Fig. 7. The hydrogen atoms which are the easiest to abstract are the secondary benzylic ones closest to the benzene ring, their C–H bond strength being $86.5 \text{ kcal mol}^{-1}$ compared to $98.5 \text{ kcal mol}^{-1}$ for the secondary $\dot{\text{H}}$ atoms and $101.1 \text{ kcal mol}^{-1}$ for the primary $\dot{\text{H}}$ atoms located progressively further from the benzene ring. Hydrogen atoms are mainly abstracted by $\dot{\text{O}}\text{H}$, $\dot{\text{H}}$, HO_2 and $\dot{\text{C}}\text{H}_3$ radicals together with molecular oxygen (O_2).

Abstraction from the innermost carbon to form a pbzjC radical is the most prominent site at this temperature and pressure. This accounts for 38.7% of the decomposition of the fuel molecule and the pbzjC radical further decomposes completely to form styrene, $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$, and methyl radicals.

The pbzjB radical is formed by abstraction of a secondary hydrogen atom on the alkyl chain, Fig. 7b. This pathway accounts for 18.0% consumption of the fuel. This fuel radical decomposes fully to form phenyl radicals and propylene.

Abstraction of a primary hydrogen atom results in the formation of a pbzjA radical, Fig. 7a. 19.1% of the fuel breaks down by this route at 1000 K. This fuel radical further decomposes fully into the benzyl radical and ethylene.

Unimolecular decomposition of the fuel into benzyl and ethyl radicals also occurs at this temperature but to a relatively small

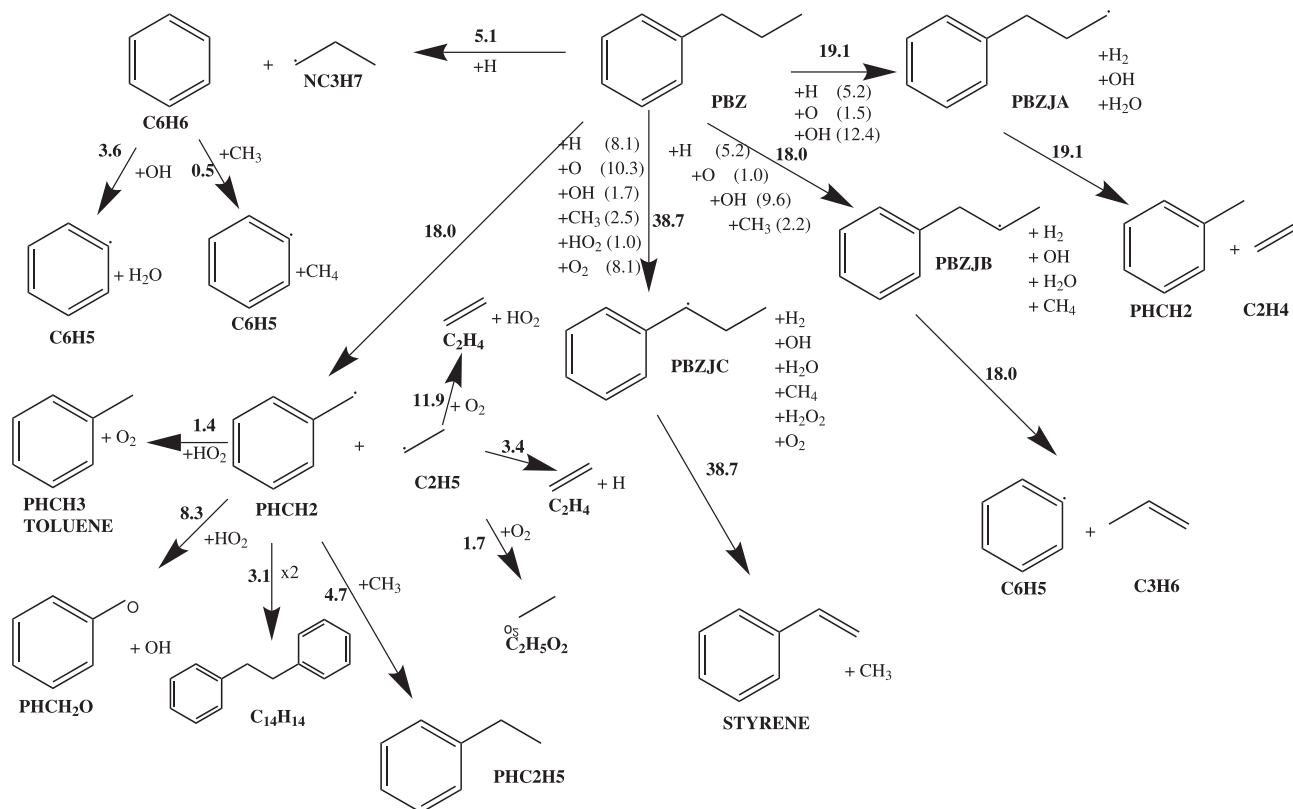


Fig. 8. Reaction path analysis for *n*-propylbenzene at 1000 K, $\phi = 0.96$, 10 atm. The numbers represent the percent of fuel flux that goes through a particular species.

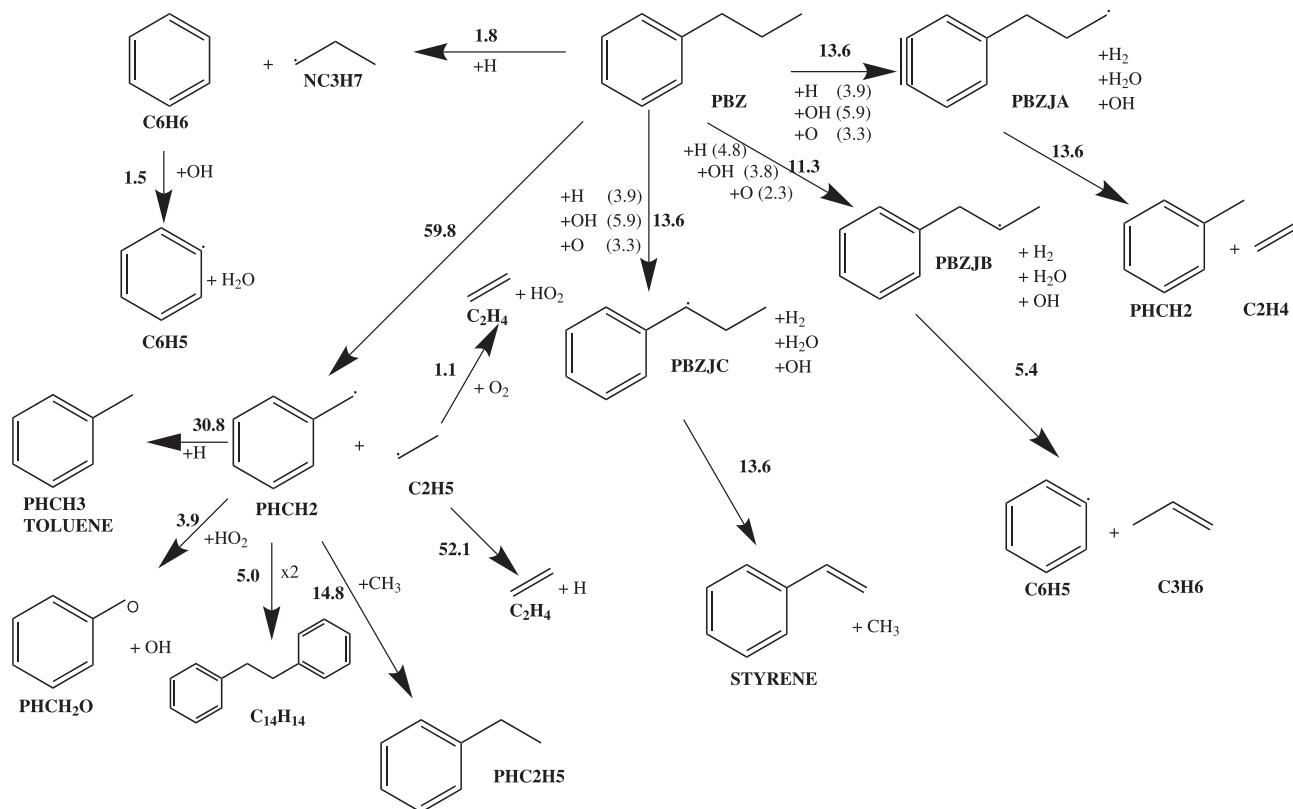


Fig. 9. Reaction path analysis for *n*-propylbenzene at 1500 K, $\phi = 0.96$, 10 atm. The numbers represent the percent of fuel flux that goes through a particular species.

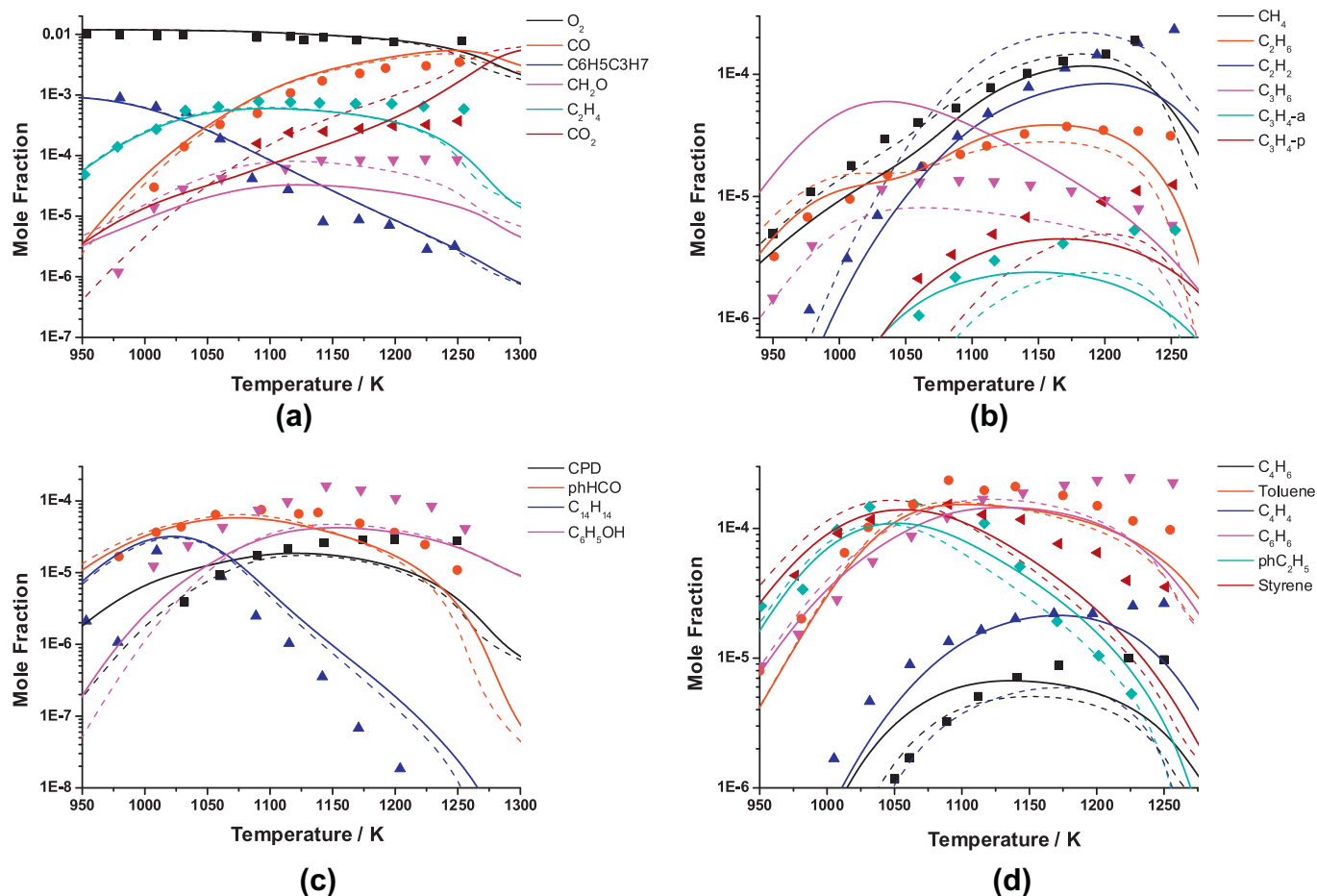


Fig. 10. *n*-Propylbenzene oxidation in a JSR at 1 atm. The initial conditions were: *n*-propylbenzene 0.1%; O₂ 1.2%, N₂ 98.7%; $\tau = 0.07$ s. Experimental data (symbols) are compared to computations (--- Diévert model, — Diévert model + unimolecular pressure dependent rates + C₀–C₄).

extent (18.0%). The ethyl radical mainly decomposes to produce ethylene and the very reactive hydrogen atom. Benzyl radicals can then be involved in one of a further four reactions. The first of which involves the combination of two benzyl radicals to form a bibenzyl molecule. The second route is the addition of a methyl radical to a benzyl radical to form ethylbenzene. The third reaction that the benzyl radical undergoes is reaction with HO₂ to form both C₆H₅CH₂Ö and ÖH radicals and the final reaction of the benzyl radical is the addition of a hydrogen atom to the radical site to form toluene and molecular oxygen.

The final reaction occurring at 1000 K is the addition of a hydrogen atom at the side chain site of the propyl group on the benzene ring to form benzene and an *n*-propyl (nC₃H₇) radical. This reaction consumes 5.1% of the fuel at this temperature, Fig. 8. Hydrogen atom abstraction from benzene by ÖH radicals results in the formation of a phenyl radical and water. Benzene also can undergo hydrogen atom abstraction by methyl radicals to form a phenyl radical and methane. *n*-Propyl radical undergoes β -scission to form ethylene and a methyl radical.

3.4.2.2. High temperature (1500 K). At 1500 K the chemistry is controlled predominantly by high temperature kinetics as the reaction pathway involving the unimolecular decomposition of the fuel becomes the dominant route in the decomposition of the fuel. At this temperature 57.7% of the fuel decomposes via this route. This is because, at higher temperatures there is enough energy in the system to cleave the C–C bond (74.8 kcal mol^{−1}) without the need for abstraction by a secondary radical. The increase in unimolecular

fuel decomposition has an increasing effect on the overall reactivity of the system as each of the products further decompose to form more reactive radicals (H, HO₂, ÖH) than formed by the decomposition of pbzjA, pbzjB and pbzjC radicals. This information along with the sensitivity analyses indicates that the reaction pathway which increases the reactivity at higher temperatures are the unimolecular fuel decompositions. These reactions occur more commonly at high temperatures (57.7% at 1500 K) as opposed to the low temperatures (18.0% at 1000 K), Fig. 9.

Unimolecular fuel decomposition occurs at the expense of hydrogen atom abstraction reactions which make a significant contribution to fuel consumption at lower temperatures but which become less important as the temperature increases. The formation of the primary fuel radicals pbzjA, pbzjB and pbzjC decrease to 13.1%, 10.9% and 13.1% respectively at 1500 K, compared to 19.1%, 18.0% and 38.7% respectively at 1000 K.

The addition of a hydrogen atom to form benzene and *n*-propyl (nC₃H₇) radical only consumes 1.7% of the fuel at 1500 K compared to 5.1% at 1000 K, Fig. 9.

Similar products are formed from the further decomposition of all these reactions as the ones mentioned above.

From both the sensitivity analysis and reaction pathway analysis described above it is clear that temperature plays a key role in the reactivity of *n*-propylbenzene as at higher temperatures unimolecular decomposition of the fuel results in increased reactivity while lower temperature products are a result of hydrogen atom abstraction reactions from the fuel resulting in less reactive products. Essentially, at lower temperatures the fuel reacts as:

Fuel + $\dot{R} = \dot{R}' + RH$, which is a chain propagation reaction, while at higher temperatures the fuel decomposes as per: Fuel = $\dot{R} + \dot{R}'$ which is a chain branching reaction as it produces two radicals from one fuel molecule.

3.4.3. Comparison with previous data

Simulations were carried out on the revised *n*-propylbenzene model to compare it with data obtained by Dagaut et. al. [27] in a jet-stirred reactor. This was done to ensure the revised model continued to capture the original successfully. The results of these simulations can be found in the [Supplementary Material](#). Results obtained showed the integrity of the model was maintained. It is interesting to note that in the original Diévert model isomerization of the $C_6H_5C_3H_6$ -B, pbzjB leads to the formation of a 2-phenyl-1-propyl radical, BPHROPY which decompose further to form styrene and methyl radicals which are the same products as those formed from the decomposition of 1-phenyl-2-propyl radicals, $C_6H_5C_3H_6$ -C (pbzjC). Since we have increased the importance of the abstraction of secondary allylic hydrogen atoms forming much higher pbzjC radicals in our model, we maintain the good simulation of the styrene concentrations measured in the experiments. The reaction

producing the 2-phenyl-1-propyl radical (R1967) is still present in the kinetic mechanism but due to the changes outlined above this reaction occurs to such a small extent it is no longer detected in the reaction path analyses (see Fig. 10).

3.5. Comparison with *n*-butylbenzene data

A direct comparison was performed in order to gain insight into differences in reactivity between the closely related fuels *n*-propylbenzene and *n*-butylbenzene under the same experimental conditions. The percentage composition of both *n*-propylbenzene and *n*-butylbenzene respectively can be seen in [Tables 1 and 2](#). The

Table 2
Mixture compositions of *n*-butylbenzene mixtures studied.

ϕ	% Fuel	% O ₂	% N ₂
2.0	3.02	20.37	76.62
1.0	1.53	20.68	77.79
0.5	0.77	20.84	78.39
0.3	0.46	20.90	78.63

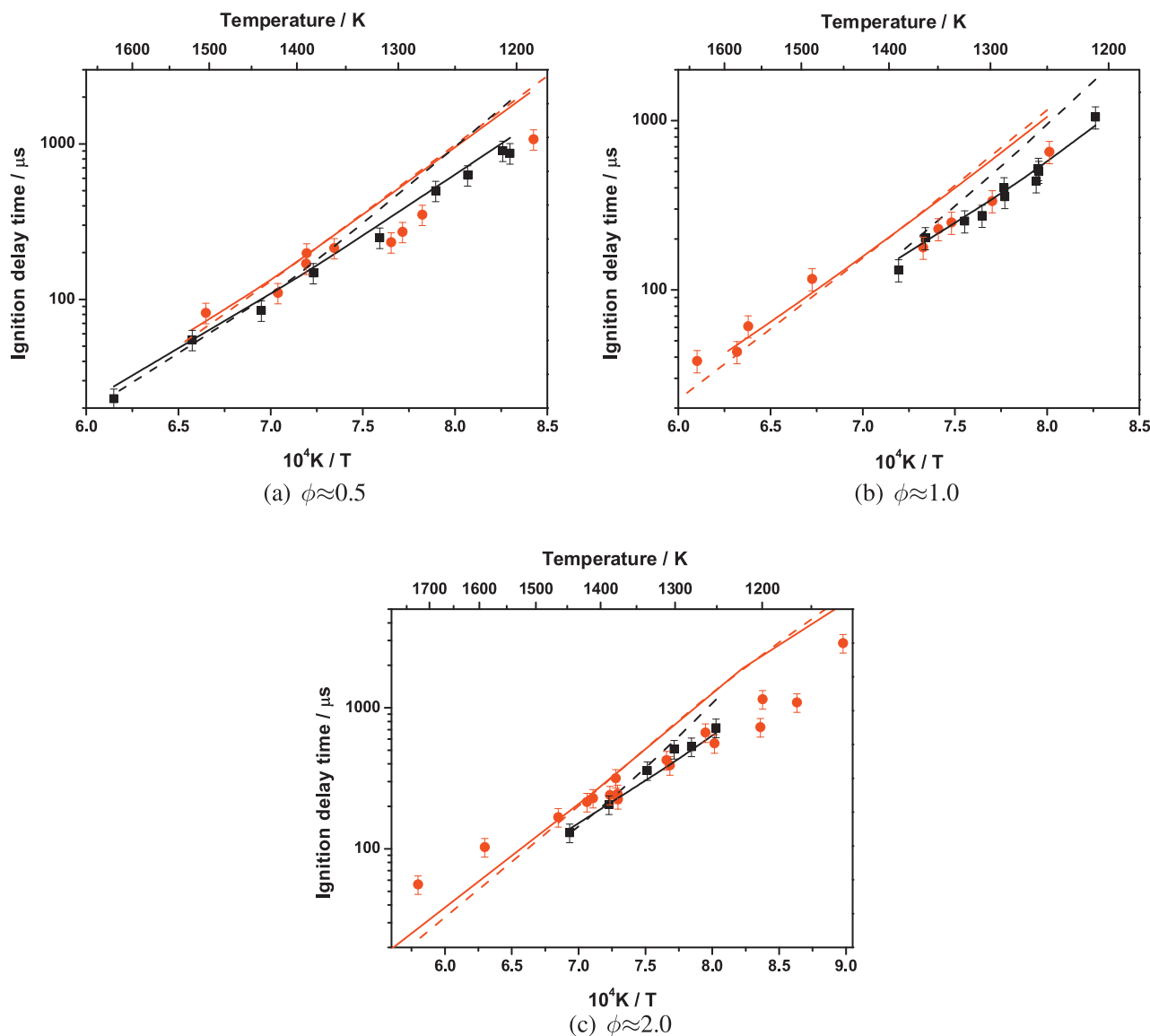


Fig. 11. A comparison of equivalence ratios (ϕ) at 1 atm. ■ – *n*-Propylbenzene data, ● – *n*-butylbenzene data. — final model, - - - original model.

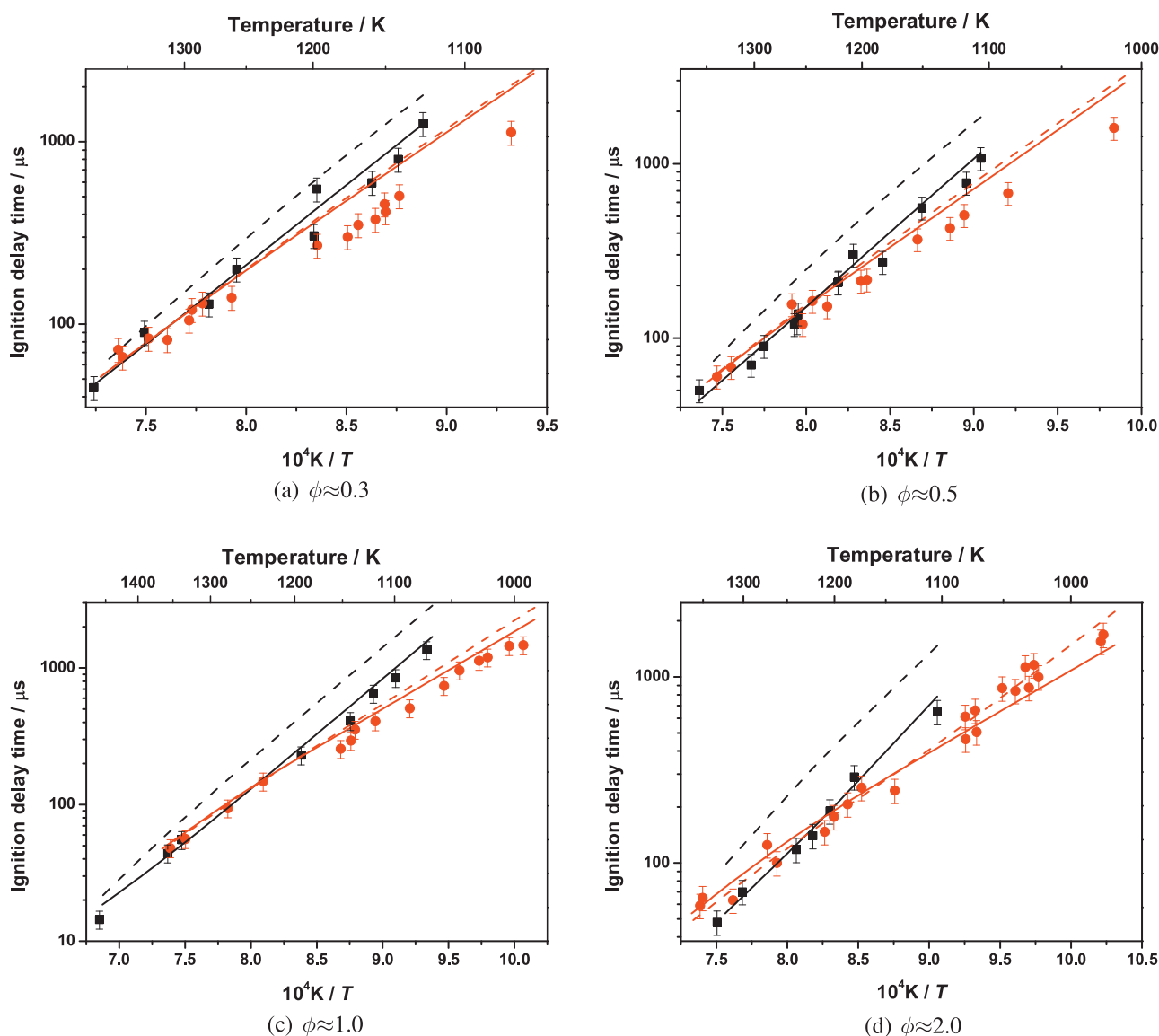


Fig. 12. A comparison of equivalence ratios (ϕ) at 10 atm. ■ – *n*-Propylbenzene data, ● – *n*-butylbenzene data. — final model, - - - original model.

resulting comparison gave an insight into the extent of reactivity differences between both closely related fuels at high temperatures. The experimental data discussed in this comparison has been previously published in a paper discussing *n*-butylbenzene oxidation [32].

3.6. Modifications to the *n*-butylbenzene model

In simulating our *n*-butylbenzene results we have used the same model as presented by Diévar et al. [29] which contains our modified C₀–C₄ chemistry and the modified *n*-propylbenzene sub-model but we have further modified the *n*-butylbenzene sub-model as described below. The addition of the C₀–C₄ sub-model, as mentioned above in the section on *n*-propylbenzene, had an effect on the slope at all pressures and equivalence ratios.

3.6.1. Unimolecular fuel decomposition

The model did not contain any pressure dependence for the unimolecular decomposition of *n*-butylbenzene. The addition of the unimolecular pressure dependence is important over all pressures studied as a large proportion of the fuel is consumed via the

decomposition of *n*-butylbenzene producing benzyl radical and $\dot{\text{C}}\text{H}_3$ which is particularly important at 1 atm where 84% of the fuel is consumed in this way.

The rate constants for unimolecular decomposition reactions were calculated as described previously for *n*-propylbenzene using the same estimate for the radical–radical recombination.

3.6.2. Hydrogen atom abstraction

Hydrogen atom abstraction from *n*-butylbenzene by a $\dot{\text{H}}$ atom had an impact on all equivalence ratios. Although very little of the fuel is consumed via this abstraction channel at 1 atm the reaction does inhibit reactivity. The rate of reaction for abstraction leading to the production of both B- and C-C₆H₅ $\dot{\text{C}}\text{H}_8$ and H₂ were decreased by a factor of two which significantly increased the reactivity at 1 atm as desired.

The rates for hydrogen atom abstraction reactions from the fuel by HO₂, $\dot{\text{H}}$, $\dot{\text{O}}$, $\dot{\text{OH}}$ radicals and by O₂ were changed to the same as those stated for *n*-propylbenzene due to the similarity of the two fuels. This resulted in a positive effect at all pressures in which the reactivity was decreased in the high temperature range while increased at lower temperatures.

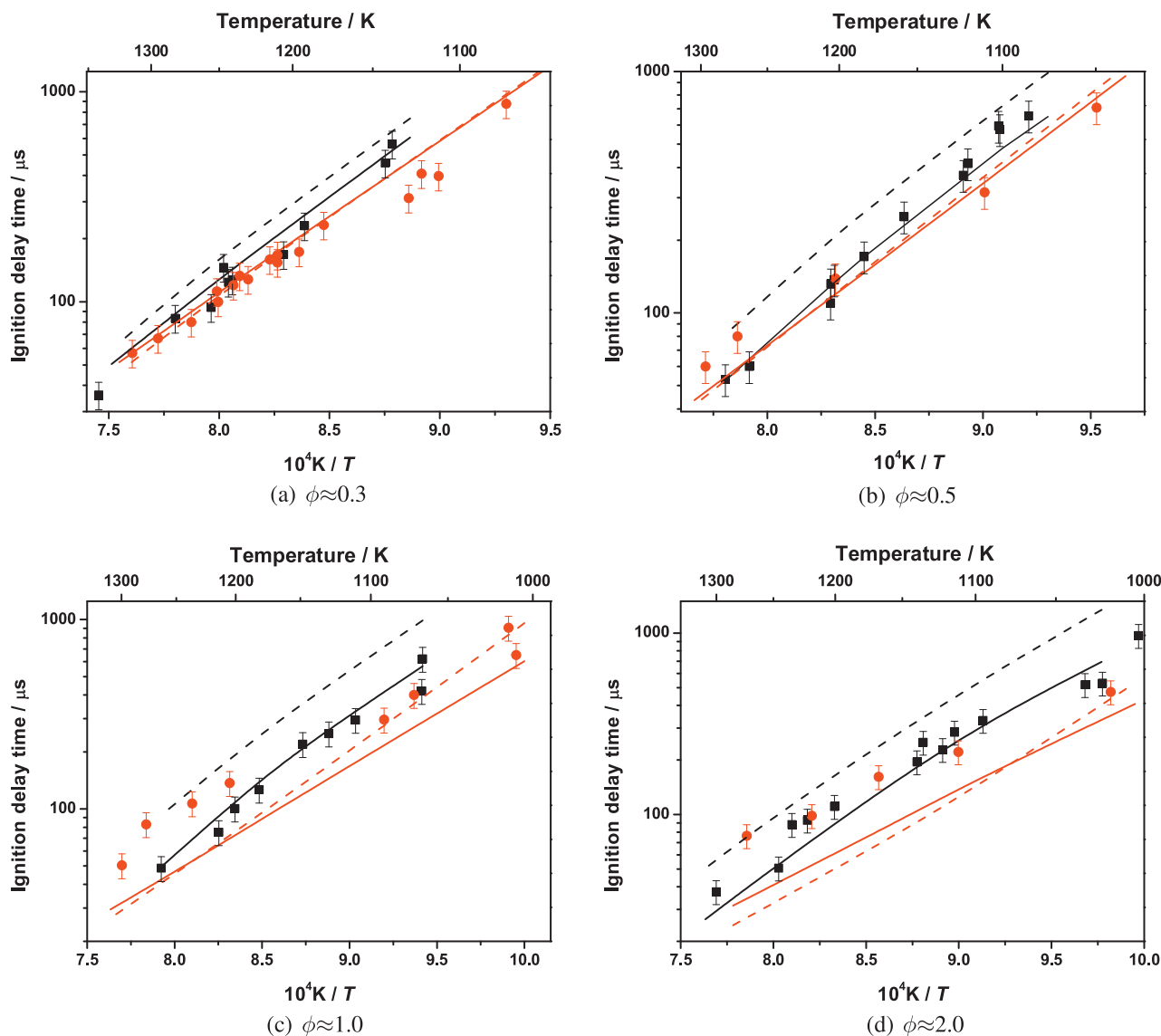


Fig. 13. A comparison of equivalence ratios (ϕ) at 30 atm. ■ – *n*-Propylbenzene data, ● – *n*-butylbenzene data. — final model, - - - original model.

In the case of hydrogen atom abstraction from the fuel by HO_2 the three reactions producing B-, C-, and D- $\text{C}_6\text{H}_5\dot{\text{C}}_4\text{H}_8$ radicals and H_2O_2 were reduced by a factor of two to reduce the reactivity at 10 and 30 atm in the low temperature region. This change had a positive effect on 10 and 30 atm simulations because after making the various desirable changes to the model to improve 1 atm simulations this lead to an increase in reactivity of both 10 and 30 atm which did not correlate with the experimental data. Upon further analyses of the rates of hydrogen atom abstraction by OH radical producing both B- and C- $\text{C}_6\text{H}_5\dot{\text{C}}_4\text{H}_8$ and H_2O the rates were increased by a factor of two thus decreasing reactivity which improved agreement between the model and experimental data at 10 and 30 atm.

3.6.3. Other rate changes

In a further attempt to increase the reactivity so that the model better reproduced the experimental data the H addition reaction to *n*-butylbenzene forming benzene and an *n*-butyl radical was increased by a factor of two. The activation energy for the recombination reaction of C_3H_6 and a benzyl radical forming B- $\text{C}_6\text{H}_5\dot{\text{C}}_4\text{H}_8$

was changed to $7800 \text{ cal mol}^{-1}$ similar to the activation energy used for methyl addition to olefins [47].

The original rate for hydrogen atom abstraction from C- $\text{C}_6\text{H}_5\dot{\text{C}}_4\text{H}_8$ by molecular oxygen forming $\text{C}_6\text{H}_5\dot{\text{C}}_4\text{H}_7\text{-3}$ radical and HO_2 radical was significantly faster than that of the same abstraction reactions at the A, B, and D positions on the butyl side chain. Therefore all the rates for hydrogen atom abstraction by molecular oxygen from the A-, B-, and D- positions on $\text{C}_6\text{H}_5\dot{\text{C}}_4\text{H}_8$ were assigned the same rate as that for C- $\text{C}_6\text{H}_5\dot{\text{C}}_4\text{H}_8$.

The reaction of A- $\text{C}_6\text{H}_5\dot{\text{C}}_2\text{H}_4$ with C_2H_4 was estimated to be $6.6 \times 10^3 T^{2.48} \exp(-3085/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ based on alkyl radical addition to ethylene as recommended by Curran [47].

3.6.4. 1 atm pressure

At a pressure of 1 atm a consistent pattern of reactivity is observed at all three equivalence ratios, Fig. 11. Both *n*-propylbenzene and *n*-butylbenzene overlap with each other at equivalence ratios of 0.5, 1.0, and 2.0 over the temperature range 1165–1750 K. This indicates that at 1 atm the reactivity of both of these fuels is similar. This indicates that at higher temperatures these

closely related fuels have very similar reactivity and as such the length of the alkyl side chain has no real effect on the reactivity.

3.6.5. 10 atm pressure

At a compressed pressure of 10 atm *n*-butylbenzene is consistently more reactive than *n*-propylbenzene in the low temperature region with the experimental data crossing over from low to high temperature chemistry at approximately 1315 K, Fig. 12. After the crossover into high temperature chemistry both fuel components exhibit similar reactivity which is consistent with the 1 atm pattern of reactivity which was taken over a higher temperature

range. At 10 atm the temperature range has shifted to a more intermediate temperature range than experienced at 1 atm. This explains that, while both fuel mixtures had similar compositions at the varying equivalence ratios, *n*-butylbenzene has a longer alkyl side chain which is important at intermediate temperatures which is dominated by chain branching and chain propagation reactions. The alkyl group dominates in terms of reactivity because of the aromatic groups stability with the butyl side chain able to undergo a wider range of hydrogen abstractions and thus create more radicals to drive reactivity at lower temperatures in comparison to the smaller propyl chain of *n*-propylbenzene.

Table 3

Elementary rate constant expressions for *n*-propylbenzene reactions modified and employed in this study. Rate constants in the form, $A T^n \exp(-E_a/RT)$, in cm³, mol, cal, and K. k_∞ relates to the rate constant for the high pressure limit, k_0 relates to the rate constant for the low pressure limit.

No	Reaction	A	n	E _a
(1916)	C ₆ H ₅ C ₃ H ₇ (+M) = C ₆ H ₅ CH ₂ + C ₂ H ₅ (+M)	$k_\infty 2.304 \times 10^{22}$ $k_0 9.9950 \times 10^{74}$	−1.586 −17.210	77,244 59,050
Troe parameters:	0.00008, 213.6, 213.6, 3565.3			
(1917)	C ₆ H ₅ C ₃ H ₇ (+M) = AC ₆ H ₅ CH ₂ H ₄ + CH ₃ (+M)	$k_\infty 7.06 \times 10^{20}$ $k_0 3.6320 \times 10^{47}$	−1.268 −9.310	89,013 68,670
Troe parameters:	0.0799, 442.17, 1.55 × 10 ¹⁰ , 2773.			
(1918)	C ₆ H ₅ C ₃ H ₇ (+M) = C ₆ H ₅ + nC ₃ H ₇ (+M)	$k_\infty 1.256 \times 10^{22}$ $k_0 6.709 \times 10^{41}$	−1.318 −7.340	102,320 84,810
Troe parameters:	0.2566, 0.054, 3.93 × 10 ¹² , 3217.3			
(1924)	C ₆ H ₅ C ₃ H ₇ + O ₂ = C ₆ H ₅ CH ₂ H ₆ -C + HO ₂	2.000×10^{13}	0.00	37,190
(1928)	C ₆ H ₅ C ₃ H ₇ + H = C ₆ H ₅ CH ₂ H ₆ -C + H ₂	3.376×10^{05}	2.36	207
(1931)	C ₆ H ₅ C ₃ H ₇ + OH = C ₆ H ₅ CH ₂ H ₆ -C + H ₂ O	2.764×10^{04}	2.64	−1919
(1937)	C ₆ H ₅ C ₃ H ₇ + HO ₂ = C ₆ H ₅ CH ₂ H ₆ -C + H ₂ O ₂	4.820×10^{03}	2.55	10,530
(1940)	C ₆ H ₅ C ₃ H ₇ + CH ₃ = C ₆ H ₅ CH ₂ H ₆ -C + CH ₄	3.690×10^{00}	3.31	4002
(1967)	C ₆ H ₅ C ₃ H ₇ + CH ₃ O = C ₆ H ₅ CH ₂ H ₆ -C + CH ₃ OH	4.000×10^{01}	2.90	8609

Table 4

Elementary rate constant expressions for *n*-butylbenzene reactions modified and employed in this study. Rate constants in the form, $A T^n \exp(-E_a/RT)$, in cm³, mol, cal, and K. k_∞ relates to the rate constant for the high pressure limit, k_0 relates to the rate constant for the low pressure limit.

No	Reaction	A	n	E _a
(2047)	C ₆ H ₅ C ₄ H ₉ (+M) = C ₆ H ₅ CH ₂ CH ₂ CH ₃	$k_\infty 3.293 \times 10^{21}$ $k_0 1.59 \times 10^{34}$	−1.593 −5.63	88,922 54,300
Troe parameters:	0.0241, 350.84, 1.19 × 10 ¹³ , 2317.10			
(2048)	C ₆ H ₅ C ₄ H ₉ (+M) = C ₆ H ₅ CH ₂ + nC ₃ H ₇	$k_\infty 2.304 \times 10^{22}$ $k_0 2.31 \times 10^{75}$	−1.586 −17.23	77,244 59,090
Troe parameters:	0.0030, 161.98, 171.13, 3417.50			
(2049)	C ₆ H ₅ C ₄ H ₉ (+M) = C ₆ H ₅ + pC ₄ H ₉	$k_\infty 2.543 \times 10^{20}$ $k_0 1.27 \times 10^{20}$	−1.239 −1.52	102,520 67,180
Troe parameters:	0.1896, 442.79, 1.55 × 10 ¹⁰ , 2734.20			
(2050)	C ₆ H ₅ C ₄ H ₉ (+M) = AC ₆ H ₅ CH ₂ H ₄ + C ₂ H ₅	$k_\infty 1.14 \times 10^{24}$ $k_0 3.64 \times 10^{38}$	−2.218 −6.79	87,260 52,480
Troe parameters:	0.0030, 337.69, 7.79 × 10 ¹² , 2258.00			
(2069)	C ₆ H ₅ C ₄ H ₉ + O ₂ = D-C ₆ H ₅ CH ₂ H ₈ + HO ₂	2.000×10^{13}	0.000	37,190
(2070)	C ₆ H ₅ C ₄ H ₉ + O ₂ = C-C ₆ H ₅ CH ₂ H ₈ + HO ₂	4.000×10^{13}	0.000	49,500
(2071)	C ₆ H ₅ C ₄ H ₉ + O ₂ = B-C ₆ H ₅ CH ₂ H ₈ + HO ₂	4.000×10^{13}	0.000	49,500
(2072)	C ₆ H ₅ C ₄ H ₉ + O ₂ = A-C ₆ H ₅ CH ₂ H ₈ + HO ₂	4.000×10^{13}	0.000	50,600
(2073)	C ₆ H ₅ C ₄ H ₉ + H = C ₆ H ₆ + pC ₄ H ₉	9.600×10^{13}	0.00	5123
(2076)	C ₆ H ₅ C ₄ H ₉ + OH = B-C ₆ H ₅ CH ₂ H ₈ + H ₂ O	1.416×10^{07}	1.90	−159
(2077)	C ₆ H ₅ C ₄ H ₉ + OH = C-C ₆ H ₅ CH ₂ H ₈ + H ₂ O	1.416×10^{07}	1.90	−159
(2078)	C ₆ H ₅ C ₄ H ₉ + OH = D-C ₆ H ₅ CH ₂ H ₈ + H ₂ O	2.764×10^{04}	2.64	−1919
(2079)	C ₆ H ₅ C ₄ H ₉ + HO ₂ = A-C ₆ H ₅ CH ₂ H ₈ + H ₂ O ₂	4.760×10^{04}	2.55	16,494
(2080)	C ₆ H ₅ C ₄ H ₉ + HO ₂ = B-C ₆ H ₅ CH ₂ H ₈ + H ₂ O ₂	4.820×10^{03}	2.60	13,910
(2081)	C ₆ H ₅ C ₄ H ₉ + HO ₂ = C-C ₆ H ₅ CH ₂ H ₈ + H ₂ O ₂	4.820×10^{03}	2.60	13,910
(2082)	C ₆ H ₅ C ₄ H ₉ + HO ₂ = D-C ₆ H ₅ CH ₂ H ₈ + H ₂ O ₂	4.820×10^{03}	2.55	10,530
(2084)	C ₆ H ₅ C ₄ H ₉ + H = B-C ₆ H ₅ CH ₂ H ₈ + H ₂	6.500×10^{05}	2.40	4470
(2085)	C ₆ H ₅ C ₄ H ₉ + H = C-C ₆ H ₅ CH ₂ H ₈ + H ₂	6.500×10^{05}	2.40	4470
(2085)	C ₆ H ₅ C ₄ H ₉ + H = D-C ₆ H ₅ CH ₂ H ₈ + H ₂	3.376×10^{05}	2.36	207
(2087)	C ₆ H ₅ C ₄ H ₉ + O = D-C ₆ H ₅ CH ₂ H ₈ + OH	9.650×10^{04}	2.68	3716
(2088)	C ₆ H ₅ C ₄ H ₉ + O = C-C ₆ H ₅ CH ₂ H ₈ + OH	4.770×10^{04}	2.71	2106
(2089)	C ₆ H ₅ C ₄ H ₉ + O = B-C ₆ H ₅ CH ₂ H ₈ + OH	4.770×10^{04}	2.71	2106
(2090)	C ₆ H ₅ C ₄ H ₉ + O = A-C ₆ H ₅ CH ₂ H ₈ + OH	1.930×10^{05}	2.68	3716
(2094)	C ₆ H ₅ C ₄ H ₉ + CH ₃ = D-C ₆ H ₅ CH ₂ H ₈ + CH ₄	3.690×10^{00}	3.31	4002
(2114)	C ₆ H ₅ C ₄ H ₉ + CH ₃ O ₂ = D-C ₆ H ₅ CH ₂ H ₈ + CH ₃ O ₂ H	4.82×10^{03}	2.90	10,530

3.6.6. 30 atm pressure

At 30 atm *n*-butylbenzene is most reactive over the entire temperature range of 1080–1350 K, Fig. 13. The 30 atm data shows a strong connection to low temperature chemical kinetics as an increase in length of the alkyl side chain of the fuel results in a increase in reactivity. The difference in reactivity becomes less pronounced as the temperature increases which indicates that high temperature kinetics is beginning to occur. Overall the 30 atm experimental data shows a similar trend in reactivity to the 10 atm data described in detail above.

3.7. Changes made to mechanism

See Fig. 13 and Tables 3 and 4.

4. Conclusions

This work represents the first comprehensive ignition delay study of *n*-propylbenzene oxidation in a heated high-pressure shock tube over a wide range of pressure, temperature and fuel/air equivalence ratio. Ignition times were measured at temperatures in the range 1000–1600 K, at reflected shock pressures of 1, 10 and 30 atm and at equivalence ratios of 0.29, 0.48, 0.96 and 1.92.

It was found that an increase in reflected shock pressure resulted in shorter ignition delay times, that is higher reactivity for all equivalence ratios investigated, which is typical of the influence of pressure on fuel reactivity. The effect of equivalence ratio on ignition delay time depended on the temperature of the experiment, where all mixtures had similar reactivity at higher temperatures and fuel-rich mixtures were most reactive at lower temperature. A convergence in reactivity was evident in the temperature range 1219–1275 K, with the “convergence temperature” dependent on the reflected shock pressure and this temperature increases with increasing pressure.

The kinetic reaction model devised by Diévar and Dagaut [29] was also improved with the addition of pressure dependent rates for the unimolecular decomposition reactions present in the original model and the replacement of the C₀–C₄ model with one in development at NUI Galway. The simulations performed using this improved model captured the experimental data very well and as such validated the model while the experimental data gave an insight into the trends in reactivity in terms of reflected shock pressure, temperature and equivalence ratio.

A comparison was made between *n*-propylbenzene and *n*-butylbenzene data obtained over the same range of conditions. A consistent trend was observed throughout the data in which the reactivity of both fuels are very similar as the while at high temperatures but *n*-butylbenzene was shows higher reactivity than *n*-propylbenzene at lower temperatures.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2012.02.009.

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